A

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By

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Declaration

I, Chirag, student of Bachelor of Computer Applications (BCA), bearing College Roll No. 2310997063, do hereby declare that the Progress Report titled **”REVADOO Gamified Rewards Web Application”** submitted in partial fulfillment of the re-quirements for the degree of Bachelor of Computer Applications is an original work carried out by me under the guidance of the organization and the faculty of Chitkara University.

I further declare that the contents of this report have not been submitted to any other institution or university for the award of any degree or diploma. All information gathered from various sources has been duly acknowledged and all permissions required for inclusion of any confidential material have been ob-tained.

Place: Chandigarh Date:
March 12, 2026

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Chirag

Abstract

In the contemporary digital landscape, user engagement and retention are criti-cal challenges for web platforms. Gamification has emerged as a powerful strat-egy to incentivize user interaction by integrating game-design elements into non-game contexts. This progress report details the development of **REVADOO**, a gamified rewards web platform currently being undertaken at Siblicode as a Ma-jor Project (22CA015).

The primary objective of REVADOO is to allow users to "Earn while they play." The platform enables users to accumulate digital credits by engaging in various activities such as playing mini-games, completing specific daily tasks, solving logic puzzles, and viewing sponsor short links. These accumulated credits can subsequently be redeemed for coupons, real cash, or saved for a planned in-app investment/stock feature.

As part of the first project evaluation, this report outlines the foundational work completed to date. This includes a comprehensive requirement analysis, the study of existing platforms like Microsoft Rewards and Taskbucks, and the formulation of both functional and non-functional requirements. Furthermore, the report presents the initial UI/UX design concepts, hardware and software specifications, and the proposed system architecture leveraging the MERN (Mon-goDB, Express.js, React.js, Node.js) stack.

The report also details the database schema and highlights the new functional-ities successfully integrated so far, including secure JWT authentication, the core credit-earning engine, and the initial integration of mini-games and task mod-ules. The document serves as a milestone marker, demonstrating the transition from conceptualization to the initial phases of technical execution.

Contents

List of Figures

List of Tables

Chapter 1

Introduction to the Organization

1.1 Overview of Siblicode

The major project, titled REVADOO, is currently being undertaken at **Siblicode**. Siblicode is a prominent, ISO-certified software and web development company headquartered in Mohali (Tricity). The organization has established itself as a reliable technology partner for businesses seeking modern, scalable, and inno-vative digital solutions.

Operating in a highly competitive IT sector, Siblicode distinguishes itself through a commitment to quality engineering, agile development methodologies, and a customer-centric approach. The company caters to a diverse clientele, ranging from emerging startups to established enterprises, providing tailored technolog-ical interventions that solve complex business challenges.

1.2 Core Expertise

Siblicode boasts a versatile and highly skilled team of developers, designers, and project managers. The core expertise of the organization spans several modern technological domains:

- **Full-Stack Web & App Development:** The organization specializes in cre-ating end-to-end web and mobile applications using modern frameworks such as the MERN stack (MongoDB, Express.js, React.js, Node.js), enabling them to deliver robust, scalable, and highly responsive applications.
- **Unity 3D Game Development:** Siblicode has a dedicated vertical for game development, utilizing Unity 3D to create immersive, interactive, and high-performance gaming experiences across various platforms.
- **UI/UX Design:** Understanding that user experience is paramount, the com-pany places a strong emphasis on user-centered design principles. Their design team creates intuitive, aesthetically pleasing, and highly functional interfaces that enhance user engagement and retention.

1.3 Mission and Vision

The organizational mission of Siblicode is strictly dedicated to leveraging the lat-est technologies to deliver innovative, high-quality solutions that drive business growth. They aim to bridge the gap between complex technical requirements and seamless user experiences.

By undertaking this major project at Siblicode, there is a unique opportunity to apply academic knowledge to real-world industry problems, specifically fo-cusing on building scalable architectures, implementing secure authentication systems, and designing engaging user interfaces for the REVADOO platform.

Chapter 2

Project Overview & Tasks Assigned

2.1 Introduction to REVADOO

REVADOO is a gamified rewards web platform operating on the core philosophy of ”Earn while you play.” In today’s digital economy, users spend countless hours online engaging with content. REVADOO aims to monetize this attention for the user by providing a unified platform where engaging with games, puzzles, and sponsor links yields tangible rewards.

Users on the platform earn digital ”Credits” through various interactive mod-ules. These credits act as the platform’s internal currency. Once a user accumu-lates a suficient balance, they can visit the Redemption module to exchange their credits for external value, such as shopping coupons, real cash transfers, or they can opt to save them for an upcoming in-app Stock/Investment page designed to teach basic financial literacy through simulated investments.

2.1.1 Core Activities for Earning Credits

The platform provides four primary avenues for users to earn credits:

1. **Play Games:** Interactive mini-games (such as memory matching, quizzes, and number games) that reward credits upon successful completion or reach-ing specific high scores.

2. **Do Tasks:** Simple daily or weekly tasks assigned to users, such as complet-ing a profile, sharing a link, or filling out a basic survey.

3. **Puzzles:** Logic-based puzzles offering varying dificulty levels. Higher dif-ficulty puzzles yield bonus credits.

4. **Short Links:** A sponsor-driven module where users view specific web links for a predetermined duration (e.g., 10 seconds) to unlock credits.

2.2 Tasks and Subtasks Assigned

For the successful execution of this major project, the development lifecycle was divided into distinct phases. As part of the first evaluation milestone, the follow-ing specific tasks and subtasks were assigned to me by the external supervisor:

- **Requirement Analysis:** Understanding the core objective of the platform, identifying the target audience, and documenting functional and non-functional requirements.

- **UI/UX Design:** Conceptualizing the visual language of the platform, ensur-ing it appeals to the target demographic, and planning the user journey from login to reward redemption.
- **Frontend Development (React.js):** Setting up the client-side application architecture, establishing routing, and building reusable UI components us-ing React and Tailwind CSS.
- **Backend Development (Node.js + Express):** Designing the server-side logic, creating RESTful API endpoints, and implementing secure user authentica-tion protocols.
- **Database Design (MongoDB):** Structuring the NoSQL document collections to eficiently store user data, credit ledgers, transactional history, and avail-able tasks.
- **Testing & Deployment Planning:** Setting up local testing environments and preparing the deployment pipeline for cloud hosting services.

Chapter 3

Work Done Before 1st Evaluation

3.1 Overview of Progress

Since the commencement of the project up to this first evaluation phase, signif-icant foundational work has been completed. The initial weeks were heavily focused on research, planning, and architectural design, followed by the com-mencement of active coding for both the front-end and back-end systems.

3.2 Survey and Study of Existing Systems

Before designing the architecture for REVADOO, a thorough analysis of existing market players in the gamified rewards sector was conducted. Two primary plat-forms were studied:

3.2.1 Microsoft Rewards

Microsoft Rewards is a loyalty program that incentivizes users to use Microsoft products (like the Bing search engine or Xbox).

- **Strengths:** Highly integrated ecosystem, reliable tracking, and a wide vari-ety of redemption options (gift cards, charity donations).
- **Limitations identified:** It is strictly tied to the Microsoft ecosystem. Users cannot earn rewards through diverse third-party mini-games or external tasks easily.
- **Takeaway for REVADOO:** The concept of daily streaks and simple point ac-cumulation (credits) is highly effective for retention and has been adapted for our platform.

3.2.2 Taskbucks

Taskbucks is a popular mobile application in India that rewards users for com-pleting tasks like downloading apps, taking surveys, or referring friends.

- **Strengths:** Direct monetary incentives (wallet cash) and a high volume of available tasks.
- **Limitations identified:** The UI can be cluttered, and the platform lacks in-trinsic entertainment value (no mini-games or logic puzzles). It feels purely transactional.

- **Takeaway for REVADOO:** By combining the transactional tasks of Taskbucks with actual entertaining games and puzzles, REVADOO can offer a more en-gaging user experience.

3.3 Requirement Analysis Outcomes

Based on the study, the specific requirements for REVADOO were finalized.

3.3.1 Target Audience

The primary target users identified for this platform are **students and young adults**. This demographic has a high afinity for gaming, possesses free time to engage in digital tasks, and is highly motivated by micro-rewards, digital coupons, and wallet cash.

3.3.2 Functional Requirements

- The system must allow users to register and securely log in.
- The system must provide a dashboard displaying the current credit balance.
- The system must host mini-games, puzzles, and short links, accurately track-ing completion.
- The system must automatically update the user's credit ledger upon task completion.
- The system must provide a catalog of redemption options (coupons, cash out).
- The system must allow administrators to add new tasks, update coupon in-ventory, and manage users.

3.3.3 Non-Functional Requirements

- **Performance:** Fast load times are crucial to prevent user drop-off, particu-larly for mini-games.
- **Security:** Secure authentication (JWT) and encrypted passwords (bcrypt) must be implemented to prevent unauthorized access and credit manipula-tion.
- **Scalability:** The database (MongoDB) must be capable of handling rapid read/write operations as transaction logs will grow quickly with every user action.

3.4 Project Scheduling

A comprehensive project timeline was prepared to guide the development pro-cess. The current milestone (Evaluation 1) marks the completion of the planning phase, environment setup, and the deployment of core structural code. Subse-quent phases will focus on integrating complex game logic, finalizing the pay-ment/redemption gateways, and rigorous system testing.

Chapter 4

Hardware & Software Requirements

To ensure smooth development, testing, and eventual usage of the REVADOO plat-form, specific hardware and software requirements have been identified. The system is designed using modern web technologies that are lightweight for the end-user but require a standard developer environment for creation.

4.1 Hardware Requirements

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher (required for running multiple local servers, database instances, and heavy IDEs simultaneously).
- **RAM:** 8 GB minimum; 16 GB recommended to ensure smooth performance while compiling React code and running Node environments.
- **Storage:** Minimum 256 GB Solid State Drive (SSD) for fast read/write oper-ations during module installations (e.g., node_modules).
- **Network:** A stable Broadband or Wi-Fi connection with a minimum speed of 10 Mbps is required to fetch API dependencies, push code to repositories, and interact with cloud databases.

4.2 Software Requirements & Technology Stack

The project leverages the MERN stack, a popular choice for building highly scal-able, data-driven web applications.

4.2.1 Frontend Technologies

- **React.js:** A JavaScript library for building user interfaces. It allows for the creation of reusable UI components and manages the view layer of the ap-plication eficiently using a Virtual DOM.
- **Tailwind CSS:** A utility-first CSS framework used to rapidly build custom, responsive user interfaces without leaving the HTML/JSX code.
- **Axios:** A promise-based HTTP client for the browser, used to make asyn-chronous API calls to the backend.
- **React Router DOM:** Enables dynamic routing within the single-page React application without requiring page reloads.

4.2.2 Backend Technologies

- **Node.js:** An asynchronous event-driven JavaScript runtime designed to build scalable network applications. It acts as the core server environment.
- **Express.js:** A minimal and flexible Node.js web application framework that provides a robust set of features to develop web and mobile applications, specifically RESTful APIs.
- **JWT (JSON Web Tokens):** An open standard used to securely transmit in-formation between parties as a JSON object. Used extensively in REVADOO for user session management and securing API routes.
- **bcrypt:** A password-hashing function used to securely store user passwords in the database.

4.2.3 Database

- **MongoDB Atlas:** A fully managed cloud database service for MongoDB. As a NoSQL database, it stores data in flexible, JSON-like documents, making it ideal for the dynamic nature of tasks and gaming records.
- **Mongoose ODM:** An Object Data Modeling (ODM) library for MongoDB and Node.js. It manages relationships between data, provides schema valida-tion, and translates between objects in code and the representation of those objects in MongoDB.

4.2.4 Development & Deployment Tools

- **Operating System:** Windows 10/11 or Ubuntu 20.04+.
- **IDEs & Tools:** Visual Studio Code (Primary IDE), Postman (for testing back-end API endpoints), Git & GitHub (Version control), npm (Node Package Manager).
- **Web Browsers:** Google Chrome (latest version) or Mozilla Firefox for test-ing and debugging.
- **Proposed Deployment:** Vercel (for hosting the React Frontend), Render (for hosting the Node Backend), and MongoDB Atlas (for database hosting).

Chapter 5

System Design and Architecture

System design is a critical phase where the requirements analyzed in the previous steps are translated into a blueprint for constructing the software. This section provides diagrammatic representations of the REVADOO architecture, logic flow, and database schema.

5.1 Data Flow Diagrams (DFD)

Data Flow Diagrams are used to provide a graphical representation of the flow of data through the information system.

5.1.1 Context Level (Level 0) DFD

The Level 0 DFD provides a high-level view of the entire REVADOO system, show-ing it as a single process interacting with external entities.

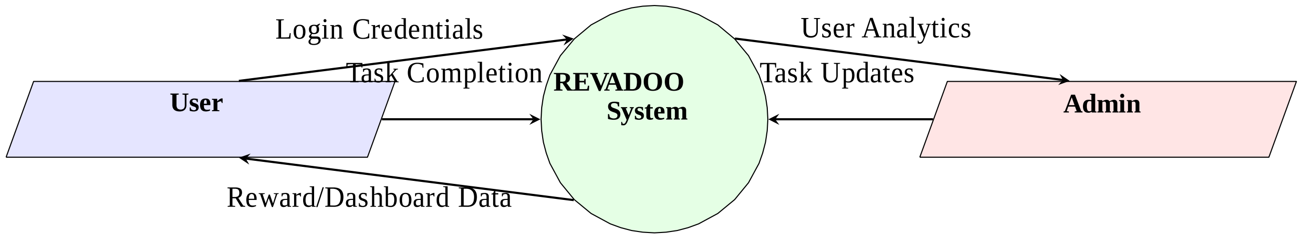

Figure 1: *Level 0 Data Flow Diagram*

5.2 System Architecture

The architecture of REVADOO follows a standard Client-Server model separated into distinct layers.

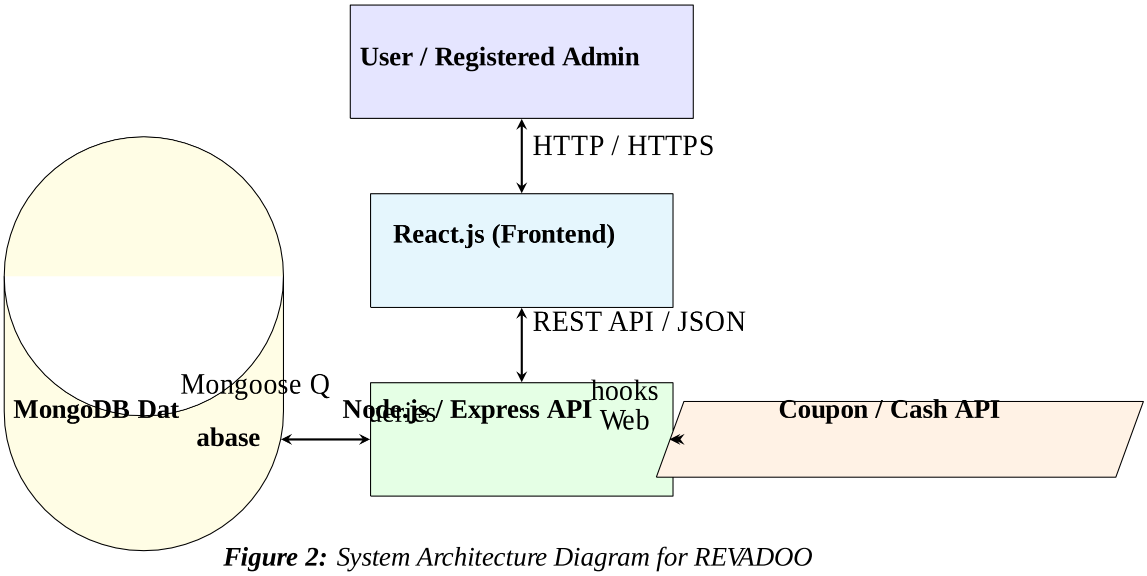

Figure 2: *System Architecture Diagram for REVADOO*

5.3 Database Schema Design

REVADOO uses MongoDB, a NoSQL database. Data is stored in collections of JSON-like documents. The schema design ensures relational integrity through object referencing despite being a NoSQL environment.

5.3.1 Collection Structures

Schema Relationships: The database follows a one-to-one (1:1) relationship be-tween USERS and CREDITS, ensuring every user has exactly one wallet. There is a one-to-many (1:N) relationship between USERS and TRANSACTIONS, as a single user will generate an ongoing log of earning and spending events over time.

Table 1: *Database Schema Definitions*

Field Name	**Data Type**	**Description**
	USERS Collection	
_id	ObjectId	Primary Key (Auto-generated)
username	String	Required, unique handle Required,
email	String	unique email address bcrypt hashed
password	String	password
role	String	’user’ or ’admin’
createdAt	Date	Account creation timestamp
	CREDITS Collection	
_id	ObjectId	Primary Key
userId	ObjectId	Foreign Key referencing USERS
balance	Number	Current available credits Lifetime
totalEarned	Number	credits earned Timestamp of last
updatedAt	Date	transaction
	TRANSACTIONS Collection	
_id	ObjectId	Primary Key
userId	ObjectId	Foreign Key referencing USERS
type	String	’earn’ or ’redeem’
amount	Number	Credit amount involved e.g.,
source	String	’puzzle_id’, ’coupon_id’
date	Date	Timestamp of transaction
	GAMES/TASKS Collection	
_id	ObjectId	Primary Key
title	String	Name of the activity ’game’,
type	String	’task’, ’puzzle’, ’link’
creditReward	Number	Credits awarded on completion
isActive	Boolean	Visibility status on frontend
	COUPONS Collection	
_id	ObjectId	Primary Key
code	String	Unique redemption code Required
creditCost	Number	balance to purchase Real-world
value expiry	Number	cash/discount value Validity period
isRedeemed	Date	Status tracking flag
	Boolean	

Chapter 6

Detailed Description of New Functionalities

During the period leading up to the first evaluation, several core functionalities and sub-modules were actively developed and integrated into the prototype. The focus has been on establishing a secure environment and laying the groundwork for the gamification engine.

6.1 User Authentication (JWT)

A robust and secure authentication system is the backbone of the platform, as it protects user credit balances.

- **Implementation:** The system uses JSON Web Tokens (JWT). When a user successfully registers or logs in, the Node.js server generates a cryptograph-ically signed token.
- **Security measures:** Passwords are not stored in plain text. The system utilizes the bcrypt library to hash passwords with a generated salt before storing them in the MongoDB USERS collection.
- **Role-based Access:** The JWT payload includes the user's role, allowing the React frontend and Express middleware to conditionally render adminis-trative panels or restrict unauthorized access to admin-only API routes (e.g., adding new tasks).

6.2 Credit Earn Engine

The Credit Earn Engine is the core backend logic responsible for managing the economy of the platform.

- **Logic Flow:** When a client-side activity reports completion, a secure POST request is sent to the engine. The engine verifies the request (ensuring the task hasn't been completed previously if it's a one-time task) and initiates a database transaction.
- **Ledger Update:** It simultaneously increments the user's balance and totalEarned in the CREDITS collection, while inserting a new log entry into the TRANSACTIONS collection for transparency.

6.3 Mini Games Module

To fulfill the "play" aspect of the platform, the foundation for interactive HTML5/React-based mini-games has been established.

- **Current Games Integrated:** The prototype currently includes logic for three simple games: a Memory Match game, a Basic Quiz, and a Number Guessing game.
- **Reward Mechanism:** Each game has a predefined `creditReward`. Upon reaching a win state within the React component, an API call triggers the Credit Earn Engine.

6.4 Tasks and Puzzles Logic

Beyond gaming, the platform caters to users seeking analytical or straightfor-ward tasks.

- **Puzzles:** A puzzle module has been implemented that classifies logic puz-zles into 5 distinct dificulty levels. Higher dificulty puzzles are mapped to higher credit rewards to incentivize user engagement.
- **Daily Tasks:** A dynamic list of daily tasks (e.g., ”Click this link”, ”Fill out profile”) is fetched from the GAMES/TASKS collection and rendered on the user dashboard.

6.5 Credit Wallet Dashboard

The dashboard serves as the central hub for the user experience.

- **Real-Time Display:** The React frontend fetches the user's credit document and displays the current balance prominently.
- **Activity Feed:** A transaction history table renders the user's recent earn-ings and spending, providing an audit trail.
- **Gamification Elements:** A visual progress bar has been coded to show users how close they are to reaching the next reward tier (e.g., ”500 cred-its away from a $5 coupon!”).

6.6 Short Link Viewer

A specialized module was developed to handle sponsor links, which is a primary revenue generation strategy for the platform.

- **Timer Implementation:** When a user clicks a short link, they are taken to a designated view. A secure 10-second timer initiates.
- **Verification:** To prevent exploitation, the timer is monitored. Only upon the verified completion of the 10 seconds does the system unlock the ”Claim Credits” button, which fires the API request to update the user's wallet.

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